

IELTS

Workbook



Reading

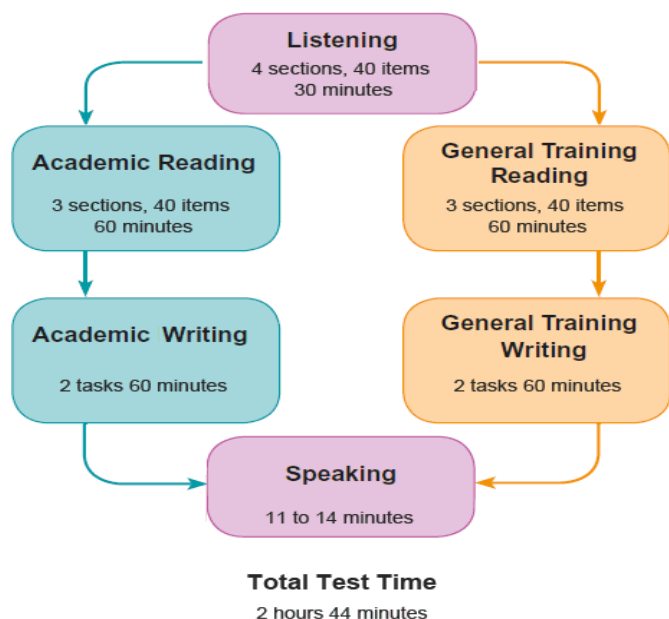
3

www.mentorsbasket.com



Preface

IELTS (International English Language Testing System) is a worldwide recognized official English exam. The IELTS is administered by the British Council, IDP: IELTS Australia and Cambridge English Language Assessment. Its aim is to evaluate the English level of the people that take it. This is one of the most accepted English exams by higher education institutions to certify English language proficiency.



IELTS Reading: Academic Module – 3 long texts of an academic nature. General Training Module - Mixture of long and short texts of a more general nature, as well as texts related to work situation.

Reading Paper is Assessed on:

- Identifying the overall purpose
- Follow key arguments in a text
- Identify opinions and attitudes
- Locate specific information
- Distinguish main ideas from details.
- Extract information from a text
- Complete a diagram, summary
- Edit a table or set of notes.

Reading Types of Questions:

- Matching Heading Questions
- True/ False/Not Given or Yes / No/ Not Given.
- Matching Paragraph Information Question.
- Summary Completion Question
- Sentence Completion Question
- Multiple Choice Question
- List Selection
- Choosing a Title
- **Categorization / Classification Question**
- Matching Sentence Endings
- Table Completion
- Flow chart Completion Question
- Short Answer Question

Such grouping questions are specially given for passages containing different types of methods, theories, processes. We need to match the statements given with the type mentioned in the question. These are quite easy to solve if the student is good at scanning the passage quickly. For such tasks you need to locate an idea or a piece of information in the text and match it to the type of event, process, method or theories which contain it. With our tips & practise papers we are sure you're going to enjoy solving such question and get correct answers.

CONTENTS

Sr. No.	Topic	Page no.
1.	Guidelines for Classification questions	4
2.	Solved Reading Passage	3 - 5
3.	Practise Reading Passage 1.1	6 - 8
4.	Practise Reading Passage 1.2	9 - 10
5.	Practise Reading Passage 1.3	11 -12
6.	Practise Reading Passage 1.4	13 - 14
7.	Practise Reading Passage 1.5	15 - 17
8.	Practise Reading Passage 1.6	18 - 19
9.	Practise Reading Passage 1.7	20 - 21
10.	Practise Reading Passage 1.8	22 - 23
11.	Practise Reading Passage 1.9	24 - 25
12.	Practise Reading Passage 1.10	26 - 28
13.	Answer key	29

TO SOLVE CLASSIFICATION QUESTIONS

Such grouping questions are specially given for passages containing different types of methods, theories, processes. We need to match the statements given with the type mentioned in the question. These are quite easy to solve if the student is good at scanning the passage quickly. For such tasks you need to locate an idea or a piece of information in the text and match it to the type of event, process, method or theories which contain it. With our tips & practise papers we are sure you're going to enjoy solving such question and get correct answers.

GUIDELINES TO SOLVE:

1. Skim read the passage to get the main idea of the passage. (5 min)
2. Understand what kind of information is given in each paragraph. You can give headings to each paragraph for your self-understanding. This way it's easy for you to locate the paragraph which contains the information asked.
3. Go through the classification question statements. Read the first statement given in the question.
4. Underline the relevant part of the statement to help you locate your answer.
5. Scan through the entire passage quickly to locate the keywords mentioned. Read the applicable part of the passage carefully. Look for the type of event or method which contains it.
6. Choose the best possible answer.
7. Spend less than 5 min to solve these questions.

Let's look at the application of the above-mentioned guidelines for the following passage below.

Solved Reading Passage

THE LITTLE ICE AGE

- A. This book will provide a detailed examination of the Little Ice Age and other climatic shifts, but, before I embark on that, let me provide a historical context. We tend to think of climate - as opposed to weather - as something unchanging, yet humanity has been at the mercy of climate change for its entire existence, with at least eight glacial episodes in the past 730,000 years. Our ancestors adapted to the universal but irregular global warming since the end of the last great Ice Age, around 10,000 years ago, with dazzling opportunism. They developed strategies for surviving harsh drought cycles, decades of heavy rainfall or unaccustomed cold; adopted agriculture and stock-raising, which revolutionised human life; and founded the world's first pre-industrial civilisations in Egypt, Mesopotamia and the Americas. But the price of sudden climate change, in famine, disease and suffering, was often high.
- B. The Little Ice Age lasted from roughly 1300 until the middle of the nineteenth century. Only two centuries ago, Europe experienced a cycle of bitterly cold winters; mountain glaciers in the Swiss Alps were the lowest in recorded memory, and pack ice surrounded Iceland for much of the year. The climatic events of the Little Ice Age did more than help shape the modern world. They are the deeply important context for the current unprecedented global warming. The Little Ice Age was far from a deep freeze, however; rather an irregular seesaw of rapid climatic shifts, few lasting more than a quarter-century, driven by complex and still little understood interactions between the atmosphere and the ocean. The seesaw brought cycles of intensely cold winters and easterly winds, then switched abruptly to years of heavy spring and early summer rains, mild winters, and frequent Atlantic storms, or to periods of droughts, light northeasterly winds, and summer heat waves.
- C. Reconstructing the climate changes of the past is extremely difficult, because systematic weather observations began only a few centuries ago, in Europe and North America. Records from India and tropical Africa are even more recent. For the time before records began, we have only 'proxy records' reconstructed largely from tree rings and ice cores, supplemented by a few incomplete written accounts. We now have hundreds of tree-ring records from throughout the northern hemisphere, and many from south of the equator, too, amplified with a growing body of temperature data from ice cores drilled in Antarctica, Greenland, the Peruvian Andes, and other locations. We are close to a knowledge of annual summer and winter temperature variations over much of the northern hemisphere going back 600 years.
- D. This book is a narrative history of climatic shifts during the past ten centuries, and some of the ways in which people in Europe adapted to them. Part One describes the Medieval Warm Period, roughly 900 to 1200. During these three centuries, Norse voyagers from Northern Europe explored northern seas, settled Greenland, and visited North America. It was not a time of uniform warmth, for then, as always since the Great Ice Age, there were constant shifts in rainfall and temperature. Mean European temperatures were about the same as today, perhaps slightly cooler.

- E. It is known that the Little Ice Age cooling began in Greenland and the Arctic in about 1200. As the Arctic ice pack spread southward, Norse voyages to the west were rerouted into the open Atlantic, then ended altogether. Storminess increased in the North Atlantic and North Sea. Colder, much wetter weather descended on Europe between 1315 and 1319, when thousands perished in a continent-wide famine. By 1400, the weather had become decidedly more unpredictable and stormier, with sudden shifts and lower temperatures that culminated in the cold decades of the late sixteenth century. Fish were a vital commodity in growing towns and cities, where food supplies were a constant concern. Dried cod and herring were already the staples of the European fish trade, but changes in water temperatures forced fishing fleets to work further offshore. The Basques, Dutch, and English developed the first offshore fishing boats adapted to a colder and stormier Atlantic. A gradual agricultural revolution in northern Europe stemmed from concerns over food supplies at a time of rising populations. The revolution involved intensive commercial farming and the growing of animal fodder on land not previously used for crops. The increased productivity from farmland made some countries self-sufficient in grain and livestock and offered effective protection against famine.
- F. Global temperatures began to rise slowly after 1850, with the beginning of the Modern Warm Period. There was a vast migration from Europe by land-hungry farmers and others, to which the famine caused by the Irish potato blight contributed, to North America, Australia, New Zealand, and southern Africa. Millions of hectares of forest and woodland fell before the newcomers' axes between 1850 and 1890, as intensive European farming methods expanded across the world. The unprecedented land clearance released vast quantities of carbon dioxide into the atmosphere, triggering for the first time humanly caused global warming. Temperatures climbed more rapidly in the twentieth century as the use of fossil fuels proliferated and greenhouse gas levels continued to soar. The rise has been even steeper since the early 1980s. The Little Ice Age has given way to a new climatic regime, marked by prolonged and steady warming. At the same time, extreme weather events like Category 5 hurricanes are becoming more frequent.

Classify the following events as occurring during the

- A. Medieval warm period
- B. Little ice age
- C. Modern warm period

Write the correct letter A, B or C

- 1. Many Europeans started farming abroad.
- 2. The cutting down of trees began to affect the climate.
- 3. Europeans discovered other lands.
- 4. Changes took place in fishing patterns.

Solved examples:

- 1. This text belongs to **Paragraph F** which mentions about the expansion of European farming methods across the world during the Modern Warm Period due to which a vast area of forest and woodland was cleared.

This can be traced from the following sentences, ‘Global Temperatures began to rise slowly after 1850, with the beginning of **Modern Warm Period**. There was a vast region..... Millions of hectares of forest and woodland fell before the newcomers’ axes between 1850 and 1890, as **intensive European farming methods expanded across the world.**’

So the above event occurred during Modern Warm Period and the correct answer is **Option C**.

- 2. This text belongs to **Paragraph F** which also describes about the human influence on climate change due to increase in cutting of trees during the **Modern Warm Period**.

This can be traced from the following sentences, ‘The unprecedented land clearance released vast quantities of carbon dioxide into the atmosphere, triggering for the first time humanly caused global warming.’

So, the above event occurred during Modern Warm Period and the correct answer is **Option C**.

Practise Reading Passage 1.1

EARLY CHILDHOOD EDUCATION

New Zealand's National Party spokesman on education, Dr Lockwood Smith, recently visited the US and Britain. Here he reports on the findings of his trip and what they could mean for New Zealand's education policy

A 'Education To Be More' was published last August. It was the report of the New Zealand Government's Early Childhood Care and Education Working Group. The report argued for enhanced equity of access and better funding for childcare and early childhood education institutions. Unquestionably, that's a real need; but since parents don't normally send children to pre-schools until the age of three, are we missing out on the most important years of all?

B A 13 year study of early childhood development at Harvard University has shown that, by the age of three, most children have the potential to understand about 1000 words - most of the language they will use in ordinary conversation for the rest of their lives.

Furthermore, research has shown that while every child is born with a natural curiosity, it can be suppressed dramatically during the second and third years of life. Researchers claim that the human personality is formed during the first two years of life, and during the first three years children learn the basic skills they will use in all their later learning both at home and at school. Once over the age of three, children continue to expand on existing knowledge of the world.

C It is generally acknowledged that young people from poorer socio-economic backgrounds tend to do less well in our education system. That's observed not just in New Zealand, but also in Australia, Britain and America. In an attempt to overcome that educational under-achievement, a nationwide programme called 'Headstart' was launched in the United States in 1965. A lot of money was poured into it. It took children into pre-school institutions at the age of three and was supposed to help the children of poorer families succeed in school.

Despite substantial funding, results have been disappointing. It is thought that there are two explanations for this. First, the programme began too late. Many children who entered it at the age of three were already behind their peers in language and measurable intelligence. Second, the parents were not involved. At the end of each day, " children returned to the same disadvantaged home environment.

D As a result of the growing research evidence of the importance of the first three years of a child's life and the disappointing results from 'Headstart', a pilot programme was launched in Missouri in the US that focused on parents as the child's first teachers. The 'Missouri' programme was predicated on research showing that working with the family, rather than bypassing the parents, is the most effective way of helping children get off to the best possible start in life. The four-year pilot study included 380 families who were about to have their first child and who represented a cross-section of socio-economic status, age and family configurations. They included single-parent and two-parent families, families in which both parents worked, and families with either the mother or father at home.

The programme involved trained parent-educators visiting the parents' home and working with the parent, or parents, and the child. Information on child development, and guidance on things to look for and expect as the child grows were provided, plus guidance in fostering the child's intellectual, language, social and motor-skill development. Periodic check-ups of the child's educational and sensory development (hearing and vision) were made to detect possible handicaps that interfere with growth and development. Medical problems were referred to professionals.

Parent-educators made personal visits to homes and monthly group meetings were held with other new parents to share experience and discuss topics of interest. Parent resource centres, located in school buildings, offered learning materials for families and facilitators for child care.

E At the age of three, the children who had been involved in the 'Missouri' programme were evaluated alongside a cross-section of children selected from the same range of socio-economic backgrounds and family situations, and also a random sample of children that age. The results were phenomenal. By the age of three, the children in the programme were significantly more advanced in language development than their peers, had made greater strides in problem solving and other intellectual skills, and were further along in social development, in fact, the average child on the programme was performing at the level of the top 15 to 20 per cent of their peers in such things as auditory comprehension, verbal ability and language ability.

Most important of all, the traditional measures of 'risk', such as parents' age and education, or whether they were a single parent, bore little or no relationship to the measures of achievement and language development. Children in the programme performed equally well regardless of socio-economic disadvantages. Child abuse was virtually eliminated. The one factor that was found to affect the child's development was family stress leading to a poor quality of parent-child interaction. That interaction was not necessarily bad in poorer families.

F These research findings are exciting. There is growing evidence in New Zealand that children from poorer socio-economic backgrounds are arriving at school less well developed and that our school system tends to perpetuate that disadvantage. The initiative outlined above could break that cycle of disadvantage. The concept of working with parents in their homes, or at their place of work, contrasts quite markedly with the report of the Early Childhood Care and Education Working Group. Their focus is on getting children and mothers access to childcare and institutionalised early childhood education. Education from the age of three to five is undoubtedly vital, but without a similar focus on parent education and on the vital importance of the first three years, some evidence indicates that it will not be enough to overcome educational inequity.

Classify the following features as characterising

- A. The ‘Headstart’ programme
- B. The ‘Missouri’ Programme
- C. Both the ‘Headstart’ and the ‘Missouri’ programmes
- D. Neither the ‘Headstart’ and the ‘Missouri’ programmes

Write the correct letter A, B, C or D

- 1. Was administered to a variety of poor and wealthy families.
- 2. Continued with follow-up assistance in elementary schools.
- 3. Did not succeed in its aim.
- 4. Supplied many forms of support and training to parents
- 5. Received insufficient funding.
- 6. Was designed to improve pre-schoolers’ educational development.

Practise Reading Passage 1.2

A CHRONICLE OF TIMEKEEPING

Our conception of time depends on the way we measure it

A According to archaeological evidence, at least 5,000 years ago, and long before the advent of the Roman Empire, the Babylonians began to measure time, introducing calendars to co-ordinate communal activities, to plan the shipment of goods and, in particular, to regulate planting and harvesting. They based their calendars on three natural cycles: the solar day, marked by the successive periods of light and darkness as the earth rotates on its axis; the lunar month, following the phases of the moon as it orbits the earth; and the solar year, defined by the changing seasons that accompany our planet's revolution around the sun.

B Before the invention of artificial light, the moon had greater social impact. And, for those living near the equator in particular, its waxing and waning was more conspicuous than the passing of the seasons. Hence, the calendars that were developed at the lower latitudes were influenced more by the lunar cycle than by the solar year. In more northern climes, however, where seasonal agriculture was practised, the solar year became more crucial. As the Roman Empire expanded northward, it organised its activity chart for the most part around the solar year.

C Centuries before the Roman Empire, the Egyptians had formulated a municipal calendar having 12 months of 30 days, with five days added to approximate the solar year. Each period of ten days was marked by the appearance of special groups of stars called decans. At the rise of the star Sirius just before sunrise, which occurred around the all-important annual flooding of the Nile, 12 decans could be seen spanning the heavens. The cosmic significance the Egyptians placed in the 12 decans led them to develop a system in which each interval of darkness (and later, each interval of daylight) was divided into a dozen equal parts. These periods became known as temporal hours because their duration varied according to the changing length of days and nights with the passing of the seasons. Summer hours were long, winter ones short; only at the spring and autumn equinoxes were the hours of daylight and darkness equal. Temporal hours, which were first adopted by the Greeks and then the Romans, who disseminated them through Europe, remained in use for more than 2,500 years.

D In order to track temporal hours during the day, inventors created sundials, which indicate time by the length or direction of the sun's shadow. The sundial's counterpart, the water clock, was designed to measure temporal hours at night. One of the first water clocks was a basin with a small hole near the bottom through which the water dripped out. The falling water level denoted the passing hour as it dipped below hour lines inscribed on the inner surface. Although these devices performed satisfactorily around the Mediterranean, they could not always be depended on in the cloudy and often freezing weather of northern Europe.

E The advent of the mechanical clock meant that although it could be adjusted to maintain temporal hours, it was naturally suited to keeping equal ones. With these, however, arose the question of when to begin counting, and so, in the early 14th century, a number of systems

evolved. The schemes that divided the day into 24 equal parts varied according to the start of the count: Italian hours began at sunset, Babylonian hours at sunrise, astronomical hours at midday and 'great clock' hours, used for some large public clocks in Germany, at midnight. Eventually these were superseded by 'small clock', or French, hours, which split the day into two 12-hour periods commencing at midnight.

F The earliest recorded weight-driven mechanical clock was built in 1283 in Bedfordshire in England. The revolutionary aspect of this new timekeeper was neither the descending weight that provided its motive force nor the gear wheels (which had been around for at least 1,300 years) that transferred the power; It was the part called the escapement. In the early 1400s came the invention of the coiled spring or fusee which maintained constant force to the gear wheels of the timekeeper despite the changing tension of its mainspring. By the 16th century, a pendulum clock had been devised, but the pendulum swung in a large arc and thus was not very efficient.

G To address this, a variation on the original escapement was invented in 1670, in England. It was called the anchor escapement, which was a lever-based device shaped like a ship's anchor. The motion of a pendulum rocks this device so that it catches and then releases each tooth of the escape wheel, in turn allowing it to turn a precise amount. Unlike the original form used in early pendulum clocks, the anchor escapement permitted the pendulum to travel in a very small arc. Moreover, this invention allowed the use of a long pendulum which could beat once a second and thus led to the development of a new floorstanding case design, which became known as the grandfather clock.

H Today, highly accurate timekeeping instruments set the beat for most electronic devices. Nearly all computers contain a quartz-crystal clock to regulate their operation. Moreover, not only do time signals beamed down from Global Positioning System satellites calibrate the functions of precision navigation equipment, they do so as well for mobile phones, instant stock-trading systems and nationwide power-distribution grids. So integral have these time-based technologies become to day-to-day existence that our dependency on them is recognised only when they fail to work.

Classify the following events as under the given list of Nationalities A-F

A. Babylonians

D. English

B. Egyptians

E. Germans

C. Greek

F. French

Write the correct letter A, B, C, D, E or F

1. They devised a civil calendar in which months were equal in length.
2. They divided the day into two equal halves.
3. They developed a new cabinet shape for a type of timekeeper.
4. They created a calendar to organise public events and work schedules

Practise Reading Passage 1.3

THE RISKS OF CIGARETTE SMOKE

Discovered in the early 1800s and named nicotianine, the oily essence now called nicotine is the main active ingredient of tobacco. Nicotine, however, is only a small component of cigarette smoke, which contains more than 4,700 chemical compounds, including 43 cancer-causing substances. In recent times, scientific research has been providing evidence that years of cigarette smoking vastly increases the risk of developing fatal medical conditions.

In addition to being responsible for more than 85 per cent of lung cancers, smoking is associated with cancers of, amongst others, the mouth, stomach and kidneys, and is thought to cause about 14 per cent of leukemia and cervical cancers. In 1990, smoking caused more than 84,000 deaths, mainly resulting from such problems as pneumonia, bronchitis and influenza. Smoking, it is believed, is responsible for 30 per cent of all deaths from cancer and clearly represents the most important preventable cause of cancer in countries like the United States today.

Passive smoking, the breathing in of the side-stream smoke from the burning of tobacco between puffs or of the smoke exhaled by a smoker, also causes a serious health risk. A report published in 1992 by the US Environmental Protection Agency (EPA) emphasized the health dangers, especially from side-stream smoke. This type of smoke contains more, smaller particles and is therefore more likely to be deposited deep in the lungs. On the basis of this report, the EPA has classified environmental tobacco smoke in the highest risk category for causing cancer.

As an illustration of the health risks, in the case of a married couple where one partner is a smoker and one a non-smoker, the latter is believed to have a 30 per cent higher risk of death from heart disease because of passive smoking. The risk of lung cancer also increases over the years of exposure and the figure jumps to 80 per cent if the spouse has been smoking four packs a day for 20 years. It has been calculated that 17 per cent of cases of lung cancer can be attributed to high levels of exposure to second-hand tobacco smoke during childhood and adolescence.

A more recent study by researchers at the University of California at San Francisco (UCSF) has shown that second-hand cigarette smoke does more harm to non-smokers than to smokers. Leaving aside the philosophical question of whether anyone should have to breathe someone else's cigarette smoke, the report suggests that the smoke experienced by many people in their daily lives is enough to produce substantial adverse effects on a person's heart and lungs.

The report, published in the Journal of the American Medical Association (AMA), was based on the researchers' own earlier research but also includes a review of studies over the past few years. The American Medical Association represents about half of all US doctors and is a strong opponent of smoking. The study suggests that people who smoke cigarettes are continually damaging their cardiovascular system, which adapts in order to compensate for the effects of smoking. It further states that people who do not smoke do not have the benefit of their system adapting to the smoke inhalation. Consequently, the effects of passive smoking are far greater on non-smokers than on smokers.

This report emphasizes that cancer is not caused by a single element in cigarette smoke; harmful effects to health are caused by many components. Carbon monoxide, for example, competes with oxygen in red blood cells and interferes with the blood's ability to deliver life giving oxygen to the heart. Nicotine and other toxins in cigarette smoke activate small blood cells called platelets, which increases the likelihood of blood clots, thereby affecting blood circulation throughout the body.

The researchers criticize the practice of some scientific consultants who work with the tobacco industry for assuming that cigarette smoke has the same impact on smokers as it does on non-smokers. They argue that those scientists are underestimating the damage done by passive smoking and, in support of their recent findings, cite some previous research which points to passive smoking as the cause for between 30,000 and 60,000 deaths from heart attacks each year in the United States. This means that passive smoking is the third most preventable cause of death after active smoking and alcohol-related diseases.

The study argues that the type of action needed against passive smoking should be similar to that being taken against illegal drugs and AIDS (SIDA). The UCSF researchers maintain that the simplest and most cost-effective action is to establish smoke-free work places, schools and public places.

Classify the following statements as being

- A. A finding of the UCSF study
- B. An opinion of the UCSF study
- C. A finding of the EPA report
- D. An assumption of the consultants to the tobacco industry

Write the correct letter A, B, C or D

1. Smoker's cardiovascular systems adapt to the intake of environmental smoke.
2. There is a philosophical question as to whether people should have to inhale others' smoke.
3. Smoke-free public places offer best solution.
4. The intake of side-stream smoke is more harmful than smoke exhaled by a smoker.

Practise Reading Passage 1.4

SYDNEY TRAVEL COLLEGE

At this College, we recommend the Multiplan policy.

Travel insurance requirements

As this course includes a total of three months' travel outside Australia, travel insurance is compulsory. If you are sick or have an accident in Australia, your medical bills will be fully covered - however, you cannot assume that everything will be covered overseas, so please read the following requirements carefully.

1. Medical

Australia has reciprocal medical arrangements with the governments of the eight nations you will be visiting. This arrangement will cover all emergency hospital treatment. However, students will have to take out insurance such as Multiplan to cover the costs of all visits to doctors, and other non-emergency medical situations.

If you have a serious accident or illness, Multiplan insurance will cover the cost of your flight back to Australia, if required. Depending on the circumstances, this may also pay for either medical personnel or a family member to accompany you home. Multiplan insurance may not cover all pre-existing medical conditions - so before you leave be sure to check with them about any long-term illnesses or disabilities that you have.

If you do require medical treatment overseas, and you want to make a claim on your insurance, the claim will not be accepted unless you produce both your student card and your travel insurance card.

2. Belongings

The Multiplan policy covers most student requirements. In particular it provides students with luggage insurance. This covers any loss or theft of your everyday belongings. For example, this insurance covers:

- the present value of items that are stolen, provided that you have purchase receipts for every item; if no receipts, no payment can be made
- replacement value of your briefcase or backpack and study books
- portable computers and CD players, if you specifically list them as items in the policy

3. Cancellation

This insurance covers any non-refundable deposit and other costs you have paid if you have to cancel due to 'unforeseen or unforeseeable circumstances outside your control'. It does not provide cover if you change your study or travel plans for other reasons.

Classify the following events as being

- A. Covered by government arrangements
- B. Covered by Multiplan policy
- C. Not covered by Multiplan policy
- D. Covered in some situations

Write the correct letter A, B, C or D

1. A student travelling overseas suddenly needs hospital treatment.
2. A student consults a doctor regarding a minor problem while abroad.
3. A parent goes overseas to bring an injured or sick student to Australia.
4. A student is treated overseas for an illness he/she had before leaving Australia.
5. A student who requires medical treatment has lost his/her travel insurance card.
6. A student's study books are lost.
7. A student's laptop is stolen.
8. A student changes his/her mind about plans to study and decides not to take the booked flight.

Practise Reading Passage 1.5

TALKING POINT

Learning a second language fuels children's intelligence and makes their job prospects brighter. But the fact is, in New Zealand, as in many other English-speaking countries, speakers of two or more languages are in the minority. Eighty-four percent of New Zealanders are monolingual (speakers of only one language). This leaves a small number who claim to speak two or more languages - a small percentage of whom were born in New Zealand.

No matter how proud people are of their cultural roots, to speak anything other than English is a marker of difference here. That's why eight-year-old Tiffany Dvorak no longer wishes to speak her mother-tongue, German, and eight-year-old Ani Powell is embarrassed when people comment on the fact that she is able to speak Maori *. As Joanne Powell, Ani's mother, points out: 'In Europe, it's not unusual for kids to be bilingual. But, if you speak another language to your children in New Zealand, there are some people who think that you are not helping them to become a member of society.'

But in fact, the general agreement among experts is that learning a second language is good for children. Experts believe that bilinguals - people who speak two languages - have a clear learning advantage over their monolingual schoolmates. This depends on how much of each language they can speak, not on which language is used, so it doesn't matter whether they are learning Maori or German or Chinese or any other language.

Cathie Elder, a professor of Language Teaching and Learning at Auckland University, says: 'A lot of studies have shown that children who speak more than one language sometimes learn one language more slowly, but in the end, they do as well as their monolingual schoolmates, and often better, in other subjects. The view is that there is an improvement in general intelligence from the effort of learning another language.'

Dr Brigitte Halford, a professor of linguistics at Freiburg University in Germany, agrees. 'Bilinguals tend to use language better as a whole,' she says. They also display greater creativity and problem-solving ability, and they learn further languages more easily.'

So with all of the benefits, why do we not show more enthusiasm for learning other languages? Parents and teachers involved in bilingual education say pressure from friends at school, general attitudes to other languages in English-speaking countries, and problems in the school system are to blame.

In New Zealand, immigrants face the possibility of culture being lost along with the language their children no longer wish to speak. Tiffany's mother, Susanne Dvorak, has experienced this. When she and husband Dieter left Germany six years ago to start up a new life in New Zealand, they thought it would be the perfect opportunity to raise their two-year-old as a bilingual. After all, bilingual Turkish families in Germany were normal and Susanne had read all the books she could find on the subject.

The idea was to have home as a German language environment and for Tiffany to learn English at nursery school. But when Tiffany went to nursery school she stopped talking completely. She was quiet for about two or three months. Then, when she took up talking again, it was only in English. Concerned for her language development, Dieter started speaking English to his daughter while Susanne continued in German.

Today, when Susanne speaks to her daughter in German, she still answers in English. ‘Or sometimes she speaks half and half. I checked with her teacher and she very seldom mixes up German and English at school. She speaks English like a New Zealander. It’s her German that’s behind,’ says Susanne.

Professor Halford, also a mother of two bilingual children, says, ‘It’s normal for kids to refuse to speak their home language at the stage when they start to socialise with other kids in kindergarten or school’. But, she says, this depends a lot on the attitudes of the societies in question. In monolingual societies, like New Zealand, ‘kids want to be like all the others and sometimes use bilingualism as one of the battlefields for finding their own identity in contrast to that of their parents.’

She supports Susanne’s approach of not pressuring her daughter. ‘Never force the child to use a specific language, just keep using it yourself. The child will accept that. There is often a time when children or teenagers will need to establish their own identity as different from their schoolmates and they may use their other language to do so.’

Cathie Elder thinks immigrant parents should only speak English to their children if they are able to use English well themselves. ‘What parents should do is provide rich language experiences for their children in whatever language they speak well. They may feel like outsiders and want to speak the local language, but it is more important for the child’s language development to provide a lot of language experience in any language.’

There can be differences between children in attitudes to learning languages. Susanne Dvorak’s two-year-old son, Danyon, is already showing signs of speaking German and English equally well. While her ‘ideal’ scenario hasn’t happened with Tiffany, she is aware that her daughter has a certain bilingual ability which, although mainly passive at this stage, may develop later on.

Joanne Powell feels the same way about her daughter, Ani. ‘At the moment she may not want to speak Maori but that’s okay because she’ll pick it up again in her own time. It’s more important that she has the ability to understand who she is. By learning another language she can open the door to another culture.’

Donna Chan, 25, a marketing specialist for IBM, arrived here with her parents from Hong Kong when she was four. She also remembers refusing to speak Chinese when she started primary school. But now she appreciates she had the chance to be bilingual. ‘It’s quite beneficial speaking another language in my job. Last year, my company sent me to a trade fair in Hong Kong because I could speak Chinese. Being bilingual definitely opens doors,’ she says.

*** Maori:** *the language spoken by the Maori people, the first native people of New Zealand.*

Classify the following statements as under the given list of people A-E

- A. Cathie Elder
- B. Brigitte Halford
- C. Susanne Dvorak
- D. Joanne Powell
- E. Donna Chan

Write the correct letter A, B, C, D or E

1. Children learning two languages may learn one language faster.
2. It has been unexpectedly difficult to raise a bilingual child in New Zealand.
3. Her daughter sometimes speaks a mixture of 2 languages.
4. Children's attitudes to language depend on general social attitudes.
5. It is not important which language provides opportunities to learn another culture.
6. Speaking second language provides work opportunities.

Practise Reading Passage 1.6

AIR POLLUTION

Part One

A Air pollution is increasingly becoming the focus of government and citizen concern around the globe. From Mexico City and New York, to Singapore and Tokyo, new solutions to this old problem are being proposed, Mailed and implemented with ever increasing speed. It is feared that unless pollution reduction measures are able to keep pace with the continued pressures of urban growth, air quality in many of the world's major cities will deteriorate beyond reason.

B Action is being taken along several fronts: through new legislation, improved enforcement and innovative technology. In Los Angeles, state regulations are forcing manufacturers to try to sell ever cleaner cars: their first of the cleanest, titled "Zero Emission Vehicles", hove to be available soon, since they are intended to make up 2 per cent of sales in 1997. Local authorities in London are campaigning to be allowed to enforce anti-pollution lows themselves; at present only the police have the power to do so, but they tend to be busy elsewhere. In Singapore, renting out toad space to users is the way of the future.

C When Britain's Royal Automobile Club monitored the exhausts of 60,000 vehicles, it found that 12 per cent of them produced more than half the total pollution. Older cars were the worst offenders; though a sizeable number of quire new cars were also identified as gross polluters, they were simply badly tuned. California has developed a scheme to get these gross polluters off the streets: they offer a flat \$700 for any old, run-down vehicle driven in by its owner. The aim is to remove the heaviest-polluting, most decrepit vehicles from the roads.

D As part of a European Union environmental programme, a London council is resting an infra-red spectrometer from the University of Denver in Colorado. It gauges the pollution from a passing vehicle - more useful than the annual stationary rest that is the British standard today - by bouncing a beam through the exhaust and measuring what gets blocked. The councils next step may be to link the system to a computerised video camera able to read number plates automatically.

E The effort to clean up cars may do little to cut pollution if nothing is done about the tendency to drive them more. Los Angeles has some of the world's cleanest cars - far better than those of Europe - but the total number of miles those cars drive continues to grow. One solution is car-pooling, an arrangement in which a number of people who share the same destination share the use of one car. However, the average number of people in o car on the freeway in , which is 1.0, has been falling steadily. Increasing it would be an effective way of reducing emissions as well as easing congestion. The trouble is, Los Angeles seem to like being alone in their cars.

F Singapore has for a while had o scheme that forces drivers to buy a badge if they wish to visit a certain part of the city. Electronic innovations make possible increasing sophistication: rates can vary according to road conditions, time of day and so on. Singapore is advancing in this direction, with a city-wide network of transmitters to collect information and charge drivers as they pass

certain points. Such road-pricing, however, can be controversial. When the local government in Cambridge, England, considered introducing Singaporean techniques, it faced vocal and ultimately successful opposition.

Part Two

The scope of the problem facing the world's cities is immense. In 1992, the United Nations Environmental Programme and the World Health Organisation (WHO) concluded that all of a sample of twenty megacities - places likely to have more than ten million inhabitants in the year 2000 - already exceeded the level the WHO deems healthy in at least one major pollutant. Two-thirds of them exceeded the guidelines for two, seven for three or more.

Of the six pollutants monitored by the WHO - carbon dioxide, nitrogen dioxide, ozone, sulphur dioxide, lead and particulate matter - it is this last category that is attracting the most attention from health researchers. PM10, a sub-category of particulate matter measuring ten-millionths of a metre across, has been implicated in thousands of deaths a year in Britain alone. Research being conducted in two counties of Southern California is reaching similarly disturbing conclusions concerning this little-understood pollutant.

A world-wide rise in allergies, particularly asthma, over the past four decades is now said to be linked with increased air pollution. The lungs and brains of children who grow up in polluted air offer further evidence of its destructive power. The old and ill, however, are the most vulnerable to the acute effects of heavily polluted stagnant air. It can actually hasten death, as it did in December 1991 when a cloud of exhaust fumes lingered over the city of London for over a week.

The United Nations has estimated that in the year 2000 there will be twenty-four mega-cities and a further eighty-five cities of more than three million people. The pressure on public officials, corporations and urban citizens to reverse established trends in air pollution is likely to grow in proportion with the growth of cities themselves. Progress is being made. The question, though, remains the same: 'Will change happen quickly enough?'

Classify the following solutions under appropriate locations given below

- | | |
|--------------|----------------|
| A. Singapore | E. Mexico city |
| B. Tokyo | F. Cambridge |
| C. London | G. Los Angeles |
| D. New York | |

Write the correct letter A, B, C, D, E, F or G

1. Manufacturers must sell cleaner cars.
2. Authorities want to have power to enforce anti-pollution laws.
3. Drivers will be charged according to the exhaust emissions.
4. Commuters are encouraged to share their vehicles with others.

Practise Reading Passage 1.7

THE DEPARTMENT OF ETHNOGRAPHY

The Department of Ethnography was created as a separate department within the British Museum in 1946, after 140 years of gradual development from the original Department of Antiquities. It is concerned with the people of Africa, the Americas, Asia, the Pacific and parts of Europe. While this includes complex kingdoms, as in Africa, and ancient empires, such as those of the Americas, the primary focus of attention in the twentieth century has been on small-scale societies. Through its collections, the Department's specific interest is to document how objects are created and used, and to understand their importance and significance to those who produce them. Such objects can include both the extraordinary and the mundane, the beautiful and the banal.

The collections of the Department of Ethnography include approximately 300,000 artefacts, of which about half are the product of the present century. The Department has a vital role to play in providing information on non-Western cultures to visitors and scholars. To this end, the collecting emphasis has often been less on individual objects than on groups of material which allow the display of a broad range of a society's cultural expressions.

Much of the more recent collecting was carried out in the field, sometimes by Museum staff working on general anthropological projects in collaboration with a wide variety of national governments and other institutions. The material collected includes great technical series - for instance, of textiles from Bolivia, Guatemala, Indonesia and at

areas of West Africa - or of artefact types such as boats. The latter include working examples of coracles from India, reed boats from Lake Titicaca in the Andes, kayaks from the Arctic, and dug-out canoes from several countries. The field assemblages, such as those from the Sudan, Madagascar and Yemen, include a whole range of material culture representative of one people. This might cover the necessities of life of an African herdsman or an Arabian farmer, ritual objects, or even on occasion airport art. Again, a series of acquisitions might represent a decade's fieldwork documenting social experience as expressed in the varieties of clothing and jewellery styles, tents and camel trappings from various Middle Eastern countries, or in the developing preferences in personal adornment and dress from Papua New Guinea. Particularly interesting are a series of collections which continue to document the evolution of ceremony and of material forms for which the Department already possesses early (if not the earliest) collections formed after the first contact with Europeans.

The importance of these acquisitions extends beyond the objects themselves. They come to the Museum with documentation of the social context, ideally including photographic records. Such acquisitions have multiple purposes. Most significantly they document for future change. Most people think of the cultures represented in the collection in terms of the absence of advanced technology. In fact, traditional practices draw on a continuing wealth of technological ingenuity. Limited resources and ecological constraints are often overcome by personal skills that would be regarded as exceptional in the West. Of growing interest is the way in which much of what we might see as disposable is, elsewhere, recycled and reused.

With the Independence of much of Asia and Africa after 1945, it was assumed that economic progress would rapidly lead to the disappearance or assimilation of many small-scale societies. Therefore, it was felt that the Museum should acquire materials representing people whose art or material culture, ritual or political structures were on the point of irrevocable change. This attitude altered with the realisation that marginal communities can survive and adapt in spite of partial integration into a notoriously fickle world economy. Since the seventeenth century, with the advent of trading companies exporting manufactured textiles to North America and Asia, the importation of cheap goods has often contributed to the destruction of local skills and indigenous markets. On the one hand modern imported goods may be used in an everyday setting, while on the other hand other traditional objects may still be required for ritually significant events. Within this context trade and exchange attitudes are inverted. What are utilitarian objects to a Westerner may be prized objects in other cultures - when transformed by local ingenuity - principally for aesthetic value. In the same way, the West imports goods from other peoples and in certain circumstances categorises them as 'art'.

Collections act as an ever-expanding database, not merely for scholars and anthropologists, but for people involved in a whole range of educational and artistic purposes. These include schools and universities as well as colleges of art and design. The provision of information about non-Western aesthetics and techniques, not just for designers and artists but for all visitors, is a growing responsibility for a Department whose own context is an increasingly multicultural European society.

Classify the following exhibits under the given list of Collection types

AT- Artefact Types

EC- Evolution of Ceremony

FA-Field Assemblages

SE- Social Experience

TS- Technical series

Write the correct letter AT, EC, FA, SE or TS

1. Bolivian textiles
2. Indian coracles
3. Airport art
4. Arctic kayaks
5. Necessities of life of an Arabian farmer
6. Tents from the middle east

Practise Reading Passage 1.8

CHILDREN'S THINKING

One of the most eminent of psychologists, Clark Hull, claimed that the essence of reasoning lies in the putting together of two 'behaviour segments' in some novel way, never actually performed before, so as to reach a goal.

Two followers of Clark Hull, Howard and Tracey Kendler, devised a test for children that was explicitly based on Clark Hull's principles. The children were given the task of learning to operate a machine so as to get a toy. In order to succeed they had to go through a two-stage sequence. The children were trained on each stage separately. The stages consisted merely of pressing the correct one of two buttons to get a marble; and of inserting the marble into a small hole to release the toy.

The Kendlers found that the children could learn the separate bits readily enough. Given the task of getting a marble by pressing the button they could get the marble; given the task of getting a toy when a marble was handed to them, they could use the marble. (All they had to do was put it in a hole.) But they did not for the most part 'integrate', to use the Kendlers' terminology. They did not press the button to get the marble and then proceed without further help to use the marble to get the toy. So the Kendlers concluded that they were incapable of deductive reasoning.

The mystery at first appears to deepen when we learn, from another psychologist, Michael Cole, and his colleagues, that adults in an African culture apparently cannot do the 'task' either. But it lessens, on the other hand, when we learn that a task was devised which was strictly analogous to the Kendlers' one but much easier for the African males to handle.

Instead of the button-pressing machine, Cole used a locked box and two differently coloured match-boxes, one of which contained a key that would open the box. Notice that there are still two behaviour segments — 'open the right match-box to get the key' and 'use the key to open the box' — so the task seems formally to be the same. But psychologically it is quite different. Now the subject is dealing not with a strange machine but with familiar meaningful objects; and it is clear to him what he is meant to do. It then turns out that the difficulty of 'integration' is greatly reduced.

Recent work by Simon Hewson is of great interest here for it shows that, for young children, too, the difficulty lies not in the inferential processes which the task demands, but in certain perplexing features of the apparatus and the procedure. When these are changed in ways which do not at all affect the inferential nature of the problem, then five-year-old children solve the problem as well as college students did in the 'own experiments'.

Hewson made two crucial changes. First, he replaced the button-pressing mechanism in the side panels by drawers in these panels which the child could open and shut. This took away the mystery from the first stage of training. Then he helped the child to understand that there was no 'magic' about the specific marble which, during the second stage of training, the experimenter handed to him so that he could pop it in the hole and get the reward.

A child understands nothing, after all, about how a marble put into a hole can open a little door. How is he to know that any other marble of similar size will do just as well? Yet he must assume that if he is to solve the problem. Hewson made the functional equivalence of different marbles clear by playing a 'swapping game' with the children. The two modifications together produced a jump in success rates from 30 per cent to 90 per cent for five-year-olds and from 35 per cent to 72.5 per cent for four-year-olds. For three-year-olds, for reasons that are still in need of clarification, no improvement—rather a slight drop in performance - resulted from the change.

We may conclude, then, that children experience very real difficulty when faced with the Kendler apparatus; but this difficulty cannot be taken as proof that they are incapable of deductive reasoning.

Classify the following descriptions as a referring

- A. Clark Hull CH
- B. Howard and Tracy Kendler HTK
- C. Micheal Cole and colleagues MC
- D. Simon Hewson SH

Write the appropriate letters A, B or C

1. Is cited as famous in the field of psychology
2. Demonstrated that the two-stage experiment involving button-pressing and inserting a marble into a hole poses problems for certain adults as well as children.
3. Devised an experiment

Practise Reading Passage 1.9

ROBOTS

Since the dawn of human ingenuity, people have devised ever more cunning tools to cope with work that is dangerous, boring, onerous, or just plain nasty. That compulsion has culminated in robotics - the science of conferring various human capabilities on machines.

A The modern world is increasingly populated by quasi-intelligent gizmos whose presence we barely notice but whose creeping ubiquity has removed much human drudgery. Our factories hum to the rhythm of robot assembly arms. Our banking is done at automated teller terminals that thank us with rote politeness for the transaction. Our subway trains are controlled by tireless robo-drivers. Our mine shafts are dug by automated moles, and our nuclear accidents - such as those at Three Mile Island and Chernobyl - are cleaned up by robotic muckers fit to withstand radiation.

Such is the scope of uses envisioned by Karel Capek, the Czech playwright who coined the term 'robot' in 1920 (the word 'robota' means 'forced labor' in Czech). As progress accelerates, the experimental becomes the exploitable at record pace.

B Other innovations promise to extend the abilities of human operators. Thanks to the incessant miniaturisation of electronics and micromechanics, there are already robot systems that can perform some kinds of brain and bone surgery with submillimeter accuracy - far greater precision than highly skilled physicians can achieve with their hands alone. At the same time, techniques of long-distance control will keep people even farther from hazard. In 1994 a ten-foot-tall NASA robotic explorer called Dante, with video-camera eyes and with spiderlike legs, scrambled over the menacing rim of an Alaskan volcano while technicians 2,000 miles away in California watched the scene by satellite and controlled Dante's descent.

C But if robots are to reach the next stage of labour-saving utility, they will have to operate with less human supervision and be able to make at least a few decisions for themselves - goals that pose a formidable challenge. 'While we know how to tell a robot to handle a specific error,' says one expert, 'we can't yet give a robot enough common sense to reliably interact with a dynamic world.' Indeed the quest for true artificial intelligence (AI) has produced very mixed results. Despite a spasm of initial optimism in the 1960s and 1970s, when it appeared that transistor circuits and microprocessors might be able to perform in the same way as the human brain by the 21st century, researchers lately have extended their forecasts by decades if not centuries.

D What they found, in attempting to model thought, is that the human brain's roughly one hundred billion neurons are much more talented - and human perception far more complicated - than previously imagined. They have built robots that can recognise the misalignment of a machine panel by a fraction of a millimeter in a controlled factory environment. But the human mind can glimpse a rapidly changing scene and immediately disregard the 98 per cent that is irrelevant, instantaneously focusing on the woodchuck at the side of a winding forest road or the single suspicious face in a tumultuous crowd. The most advanced computer systems on Earth can't approach that kind of ability, and neuroscientists still don't know quite how we do it.

E Nonetheless, as information theorists, neuroscientists, and computer experts pool their talents, they are finding ways to get some lifelike intelligence from robots. One method renounces the linear, logical structure of conventional electronic circuits in favour of the messy, ad hoc arrangement of a real brain's neurons. These 'neural networks' do not have to be programmed. They can 'teach' themselves by a system of feedback signals that reinforce electrical pathways that produced correct responses and, conversely, wipe out connections that produced errors. Eventually the net wires itself into a system that can pronounce certain words or distinguish certain shapes.

F In other areas researchers are struggling to fashion a more natural relationship between people and robots in the expectation that some day machines will take on some tasks now done by humans in, say, nursing homes. This is particularly important in Japan, where the percentage of elderly citizens is rapidly increasing. So experiments at the Science University of Tokyo have created a 'face robot' - a life-size, soft plastic model of a female head with a video camera imbedded in the left eye - as a prototype. The researchers' goal is to create robots that people feel comfortable around. They are concentrating on the face because they believe facial expressions are the most important way to transfer emotional messages. We read those messages by interpreting expressions to decide whether a person is happy, frightened, angry, or nervous. Thus the Japanese robot is designed to detect emotions in the person it is 'looking at' by sensing changes in the spatial arrangement of the person's eyes, nose, eyebrows, and mouth. It compares those configurations with a database of standard facial expressions and guesses the emotion. The robot then uses an ensemble of tiny pressure pads to adjust its plastic face into an appropriate emotional response.

G Other labs are taking a different approach, one that doesn't try to mimic human intelligence or emotions. Just as computer design has moved away from one central mainframe in favour of myriad individual workstations - and single processors have been replaced by arrays of smaller units that break a big problem into parts that are solved simultaneously - many experts are now investigating whether swarms of semi-smart robots can generate a collective intelligence that is greater than the sum of its parts. That's what beehives and ant colonies do, and several teams are betting that legions of mini-critters working together like an ant colony could be sent to explore the climate of planets or to inspect pipes in dangerous industrial situations.

Classify the following statements as under the given list of people A-C

A. Bill Gates

B. Nello Cristianini

C. Ron Chrisley

Write the correct letter A, B or C

1. An important concern for scientists is to ensure that robots do not seem frightening.
2. We have stopped trying to enable robots to perceive objects as humans do.
3. It will take considerable time for modern robots to match the ones we have created in films and books.
4. We need to enable robots to move freely before we think about trying to communicate with them.

Practise Reading Passage 1.10

SCIENCE IN SPACE

How will NASA transform the international Space Station from a building site into a cutting – edge research laboratory?

A premier, world-class laboratory in low Earth orbit. That was how the National Aeronautics and Space Administration agency (NASA) sold the International Space Station (ISS) to the US Congress in 2001. Today no one can doubt the agency's technological ambition. The most complex engineering project ever attempted has created an enormous set of interlinked modules that orbits the planet at more than 27,000 kilometres per hour. It might be travelling fast but, say critics, as a lab it is going nowhere. So far, it has gone through \$150 billion.

So where should its future priorities lie? This question was addressed at the recent 1st annual ISS research and development conference in Colorado. Among the presenters was Satoshi Iwase of Aichi Medical University in Japan who has spent several years developing an experiment that could help solve one of the key problems that humans will face in space: keeping our bodies healthy in weightlessness. One thing that physiologists have learned is that without gravity our bodies begin to lose strength, leaving astronauts with weakened bones, muscles and cardiovascular systems. To counter these effects on a long-duration mission to, say, Mars, astronauts will almost certainly need to create their own artificial gravity. This is where Iwase comes in. He leads a team designing a centrifuge for humans. In their preliminary design, an astronaut is strapped into the seat of a machine that resembles an exercise bike. Pedalling provides a workout for the astronaut's muscles and cardiovascular system, but it also causes the seat to rotate vertically around a central axis so the rider experiences artificial gravity while exercising.

The centrifuge project highlights the station's potential as a research lab. Similar machines have flown in space aboard NASA's shuttles, but they couldn't be tested for long enough to prove whether they were effective. It's been calculated that to properly assess a centrifuge's impact on human physiology, astronauts would have to ride it for 30 minutes a day for at least two months. The only way to test this is in weightlessness, and the only time we have to do that is on the space station,' says Laurence Young, a space medicine expert at the Massachusetts Institute of Technology.

There are certainly plenty of ideas for other experiments: but many projects have yet to fly. Even if the centrifuge project gets the green light, it will have to wait another five years before the station's crew can take a spin. Lengthy delays like this are one of the key challenges for NASA, according to an April 2011 report from the US National Academy of Sciences. Its authors said they were 'deeply concerned' about the state of NASA's science research, and made a number of recommendations. Besides suggesting that the agency reduces the time between approving experiments and sending them into space, it also recommended setting clearer research priorities.

NASA has already begun to take action, hiring management consultants ProOrbis to develop a plan to cut through the bureaucracy. And Congress also directed NASA to hire an independent organisation, the Centre for the Advancement of Science in Space (CASIS), to help manage the station's US lab facilities. One of CASIS's roles is to convince public and private investors that science on the station is worth the spend because judged solely by the number of papers published, the ISS certainly seems poor value: research on the station has generated about 3,100 papers since 1998. The Hubble Space Telescope, meanwhile, has produced more than 11,300 papers in just over 20 years, yet it cost less than one-tenth of the price of the space station.

Yet Mark Uhran, assistant associate administrator for the ISS, refutes the criticism that the station hasn't done any useful research. He points to progress made on a salmonella vaccine, for example. To get the ISS research back on track, CASIS has examined more than 100 previous microgravity experiments to identify promising research themes. From this, it has opted to focus on life science and medical research, and recently called for proposals for experiments on muscle wasting, osteoporosis and the immune system. The organisation also maintains that the ISS should be used to develop products with commercial application and to test those that are either close to or already on the market. Investment from outside organisations is vital, says Uhran, and a balance between academic and commercial research will help attract this.

The station needs to attract cutting-edge research, yet many scientists seem to have little idea what goes on aboard it. Jeanne DiFrancesco at ProOrbis conducted more than 200 interviews with people from organisations with potential interests in low gravity studies. Some were aware of the ISS but they didn't know what's going on up there, she says. 'Others know there's science, but they don't know what kind.'

According to Alan Stern, planetary scientist, the biggest public relations boost for the ISS may come from the privately funded space flight industry. Companies like SpaceX could help NASA and its partners when it comes to resupplying the ISS, as it suggests it can reduce launch costs by two-thirds. Virgin Atlantic's Space Ship Two or ZeroUnfinity's high-altitude balloon could also boost the space station's fortunes. They might not come close to the ISS's orbit, yet Stern believes they will revolutionise the way we, the public, see space. Soon everyone will be dreaming of interplanetary travel again, he predicts. More importantly, scientists are already queuing for seats on these low-gravity space-flight services so they can collect data during a few minutes of weightlessness. This demand for low-cost space flight could eventually lead to a service running on a more frequent basis, giving researchers the chance to test their ideas before submitting a proposal for experiments on the ISS. Getting flight experience should help them win a slot on the station, says Stern.

Classify the following opinions under the given list of people A-C

- A. Laurence Young
- B. Authors of the US Academy of Sciences report
- C. Mark Uhran
- D. Jeanne DiFrancesco

Write the correct letter A, B, C or D

1. The ISS should be available for business-related ventures.
2. There is general ignorance about what kinds of projects are possible on the ISS.
3. The process of getting accepted projects onto the ISS should be speeded up.
4. Some achievements of the ISS are underrated.
5. To properly assess new space technology, there has to be an absence of gravity.

ANSWER KEY

PASSAGE 1.1 1. B 2. D 3. A 4. B 5. D 6. C	PASSAGE 1.2 1. B 2. F 3. D 4. A	PASSAGE 1.3 1. A 2. B 3. B 4. C
PASSAGE 1.4 1. A 2. B 3. D 4. D 5. C 6. B 7. D 8. C	PASSAGE 1.5 1. A 2. C 3. C 4. B 5. A 6. D 7. E	PASSAGE 1.6 1. G 2. C 3. A 4. C 5. G
PASSAGE 1.7 1. TS 2. AT 3. FA 4. AT 5. FA 6. SE	PASSAGE 1.8 1. A 2. C 3. C 4. D 5. D 6. C 7. B 8. D	PASSAGE 1.9 1. C 2. B 3. A 4. C
PASSAGE 1.10 1. C 2. D 3. B 4. C 5. A		

